Week 4: Fluids SOLUTIONS

Discussion Questions

1 **True/True/False**

2 As higher air velocity corresponds to lower air pressure, balloons will feel a net force inwards from the surrounding air pressure.

3 a) $v_{rms} \propto \frac{1}{\sqrt{m}}$ so since Helium is lighter than Neon, it has a great root mean square speed.

b) $K = \frac{3}{2}k_BT$ for all monatomic gases so they have the same kinetic energy.

c) $p = \frac{Nk_BT}{V}$, and they have the same $\frac{k_BT}{V}$. The jar contents weigh the same but Helium particles are lighter so there are more Helion than Neon particles. Therefore, the helium pressure is greater.

4 Speed distribution – anything above the escape velocity will leave the planet, and the remaining particles will redistribute their speeds such that some new particles have a speed greater than the escape speed, and they too will leave the planet. This process repeats until there is almost no atmosphere left. Also, hotter planets will have a greater gas kinetic energy and so a greater percentage of the speed distribution curve will be above the escape velocity.

Problem Solving Questions

5 Absolute pressure = ρgh + atmospheric pressure (101 000 Pa) = 101 MPa

6 Pressure difference = $p_0(e^{-ah_2} - e^{-ah_1})$; this is the force per m^2 = 43.9 kN

7
- tension in string = (weight – buoyant force) = ($mg - \rho_F Vg$) = 2.0 N
- in lift accelerating downwards effective g = 8.0 m/s^2; this also affects the buoyancy force, as the water would "weigh" less; answer is T = 1.6 N
- in free fall situation effective g = 0; hence T = 0 N

8 Work done by pressure = $P\Delta V = 1.01 \times 10^5 \times 1.4$; or $F \times x = P \times A \times (V/A)$

9 $$\boldsymbol{v_{rms} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3k_B(40+273)}{40u}} = 442m/s}$$

10 This question requires integration. Divide the dam wall up into horizontal strips of depth Δh. Net force on one strip = $\rho gh \times w\Delta h$. In the limit this becomes $F = \rho gw \int_0^{30} h\, dh$ = 441 MN

Past Exam Questions

11ai For floating objects $\rho_o V_o g = \rho_F V_s g$, where ρ_o is the density of the object, V_o the volume of the object, ρ_F the density of the fluid, and V_s the volume submerged. Simple algebra leads to the answer (500 kg m^{-3}).

aii The same relationship can be used; the algebra is again simple. Answer is 5/8, or 62.5%.

12a Equation of continuity ($A_1v_1 = A_2v_2$). Algebra leads to the results ($2v$ and v).

b Use Bernoulli's equation at the midline of the flow: $P_1 + ½\rho v_1^2 = P_2 + ½\rho v_2^2$. This leads to $1.5\rho v^2$.

c Set the result of part (b) equal to σgh; algebra leads to the result $v = \sqrt{\frac{2sgh}{3r}}$

13a Kinetic energy K = $3/2k_BT$, thermal energy = N*K
Need to find N:
$N = \frac{PV}{k_BT} = \frac{100\times101325\times0.015}{1.38\times10^{-23}\times293} = 3.76\times10^{25}$
So $E_{th} = 3/2Nk_BT = 2.28\times10^5 J$

b

$$\lambda = \frac{1}{4\sqrt{2}\pi r^2\eta} = \frac{1}{4\sqrt{2}\pi(10^{-10})\text{^}2\times3.76\times10^{25}/0.015} = 2.24\times10^{-9}m$$